

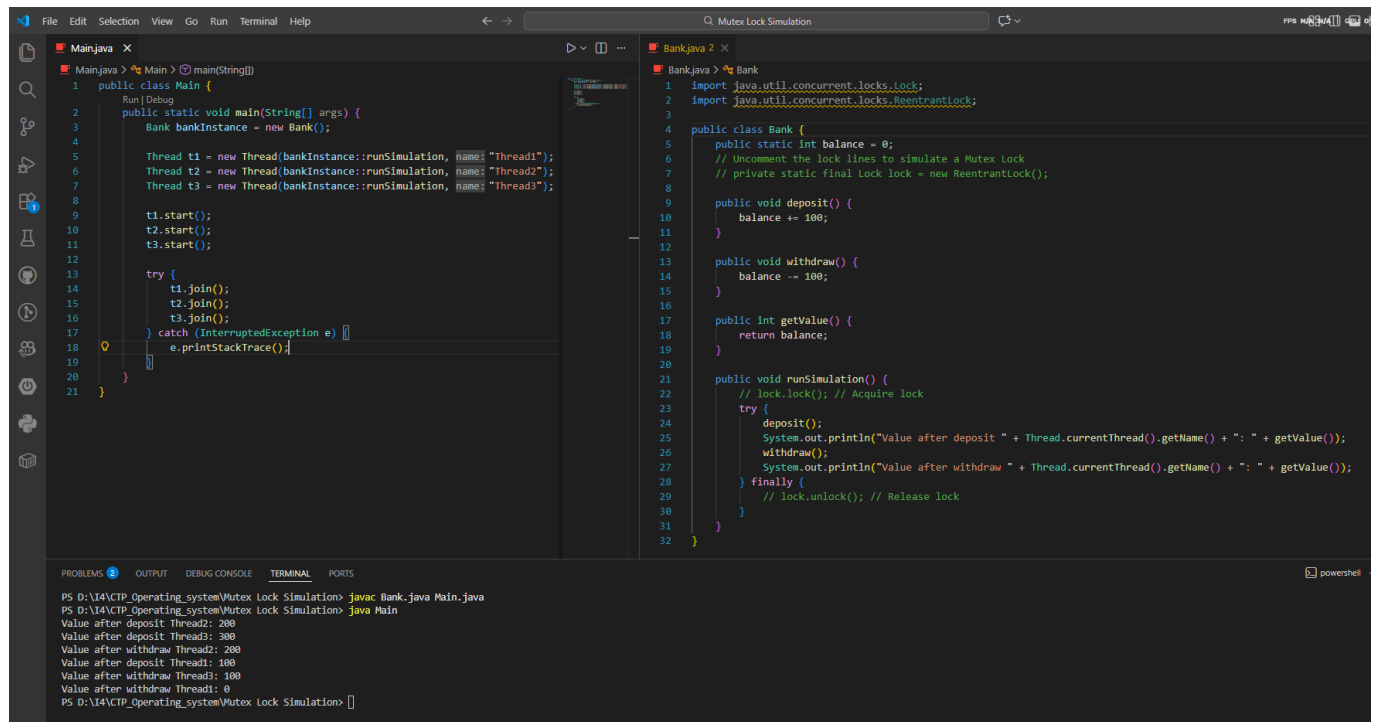
Operating System

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mutex-lock-simulation



```
1 public class Main {
2     public static void main(String[] args) {
3         Bank bankInstance = new Bank();
4
5         Thread t1 = new Thread(bankInstance::runSimulation, name: "Thread1");
6         Thread t2 = new Thread(bankInstance::runSimulation, name: "Thread2");
7         Thread t3 = new Thread(bankInstance::runSimulation, name: "Thread3");
8
9         t1.start();
10        t2.start();
11        t3.start();
12
13        try {
14            t1.join();
15            t2.join();
16            t3.join();
17        } catch (InterruptedException e) {}
18        e.printStackTrace();
19    }
20 }
21 }
```

```
1 import java.util.concurrent.locks.Lock;
2 import java.util.concurrent.locks.ReentrantLock;
3
4 public class Bank {
5     public static int balance = 0;
6     // Uncomment the lock lines to simulate a Mutex Lock
7     // private static final Lock lock = new ReentrantLock();
8
9     public void deposit() {
10        balance += 100;
11    }
12
13     public void withdraw() {
14        balance -= 100;
15    }
16
17     public int getValue() {
18        return balance;
19    }
20
21     public void runSimulation() {
22        // lock.lock(); // Acquire lock
23        try {
24            deposit();
25            System.out.println("Value after deposit " + Thread.currentThread().getName() + ": " + getValue());
26            withdraw();
27            System.out.println("Value after withdraw " + Thread.currentThread().getName() + ": " + getValue());
28        } finally {
29            // lock.unlock(); // Release lock
30        }
31    }
32 }
```

```
PS D:\I4\CTP_Operating_system\Mutex Lock Simulation> javac Bank.java Main.java
PS D:\I4\CTP_Operating_system\Mutex Lock Simulation> java Main
Value after deposit Thread2: 200
Value after deposit Thread3: 300
Value after withdraw Thread2: 200
Value after deposit Thread1: 100
Value after withdraw Thread3: 100
Value after withdraw Thread1: 0
PS D:\I4\CTP_Operating_system\Mutex Lock Simulation>
```

github link: <https://github.com/Nget-Darapich/Operating-System.git>